

# Image Captioning with a CNN Encoder and Attention-Based Decoders: A Controlled Comparison

Anthony Kang

Dept. of Electrical and Computer Engineering  
University of North Carolina at Charlotte  
Charlotte, NC, USA  
akang8@charlotte.edu

Aadit [TODO: last name]

Dept. of Electrical and Computer Engineering  
University of North Carolina at Charlotte  
Charlotte, NC, USA  
[TODO: email]

**Abstract**—We study neural image captioning on the Flickr8k dataset, using a convolutional image encoder paired with a recurrent language decoder as a controlled testbed for the effect of attention. A single frozen ResNet-50 encoder produces spatial image features that are shared, unchanged, across five decoder architectures, so that differences in caption quality primarily reflect decoder architecture under the shared training and selection protocol. Against a no-attention gated recurrent unit (GRU) baseline we compare GRU decoders with Luong (dot-product), Bahdanau (additive), and multi-head attention, and a Transformer decoder. All models are trained and selected under one protocol: a per-architecture learning-rate search, three independent random seeds, and model selection on validation BLEU-4 only, with final comparison performed through a frozen manifest after preliminary baseline pipeline verification. Contrary to the intuition that added attention should uniformly help, we find that on held-out test data Bahdanau attention achieves the highest mean BLEU-4 (0.206 vs. 0.199 for the baseline), though the baseline remains slightly higher on METEOR; dot-product (Luong) attention produced lower test BLEU-4 than the baseline under all three matched seeds (0.180), and multi-head attention and the 10M-parameter Transformer showed no consistent test-set advantage over the 4.2M-parameter baseline across the three seeds. Attention visualizations of one example illustrate a plausible failure mode for the worst model: Luong’s unscaled dot-product attention concentrates on a fixed, off-object location that ignores image content. The result is that decoder complexity does not reliably improve captioning when paired with a strong frozen encoder on a small dataset.

**Index Terms**—image captioning, attention mechanisms, recurrent neural networks, transformers, reproducibility

## I. INTRODUCTION AND MOTIVATION

Image captioning—automatically generating a natural-language description of an image—sits at the intersection of computer vision and natural language processing, the two halves of this course. A caption model must both perceive the salient content of an image and render that content as fluent text, which makes it a compact vehicle for combining a convolutional encoder with a sequence decoder.

Beyond its pedagogical value, captioning has a direct accessibility application: automatically produced “alt-text” lets screen-reading software convey visual web content to users

with visual impairments. More broadly, a model that maps images to descriptive text is a building block for weakly labeling image collections, where machine-generated descriptions can pre-annotate images before human refinement.

The central question of this project is not simply whether we can build a captioner, but *which decoder architecture works best when the visual front end is held fixed*, and by how much. Modern encoders are strong: a ResNet-50 pretrained on ImageNet already produces a rich global image representation before any language model sees it. This motivates a concrete hypothesis—that the incremental benefit of spatial attention, decisive in early captioning work built on weaker features, may be modest when the encoder is this capable and the dataset relatively small. Rather than assume attention helps, we set up an experiment in which it is free to help, tie, or lose, and we measure the result under conditions strict enough to trust. As the results show, this framing was warranted: most of the attention mechanisms we tested did not improve on a no-attention baseline once evaluated on held-out data.

Our contributions are:

- A shared, frozen ResNet-50 feature representation and a common training harness that let five decoder architectures be compared with the data, preprocessing, optimizer, and evaluation held identical.
- A reproducibility-focused protocol—content-hashed data definitions, per-architecture learning-rate search, three random seeds per model, validation-only model selection, and a manifest-gated frozen test set—that makes the comparison defensible rather than anecdotal.
- A controlled comparison of five decoders showing that additive attention achieves the highest mean test BLEU-4, together with attention visualizations that illustrate a plausible failure mode associated with dot-product attention’s lower performance.

## II. APPROACH

### A. Overview

All models share the encoder–decoder structure standard in neural captioning [7], [8]. An image is passed once through

a frozen convolutional encoder to produce spatial feature vectors; a decoder then generates a caption token by token, conditioned on those features. The *only* component that varies across our experiments is the decoder. Because the encoder is frozen and its outputs cached, every decoder is trained on numerically identical image features, which isolates architectural differences.

### B. Encoder

The encoder is a ResNet-50 [1] pretrained on ImageNet [2] (torchvision IMAGENET1K\_V2 weights). We remove the final global-average-pool and classification layers and retain the last convolutional feature map. For a  $224 \times 224$  input this yields a  $7 \times 7$  grid of 2048-dimensional vectors—49 spatial locations, each of dimension 2048.

*Note on feature-map size.* Our proposal stated a  $14 \times 14$  grid; at the  $224 \times 224$  input resolution we use, ResNet-50’s final stage produces a  $7 \times 7$  (49-location) grid. The implementation and all reported results use the correct  $7 \times 7$  features; the proposal figure was an error.

The encoder is *frozen*: its parameters receive no gradient. This is both a computational choice—each image is encoded once and cached to disk—and an experimental one, since a fixed encoder guarantees all decoders receive identical inputs.

### C. Baseline decoder (no attention)

The baseline collapses the 49 spatial vectors into a single image vector by mean-pooling, projects it through a learned linear layer with a tanh nonlinearity to initialize the hidden state of a single-layer GRU [3], and generates words autoregressively. After initialization the decoder never attends back to the image: all visual information must pass through the fixed-size initial hidden state. This bottleneck is precisely the limitation attention was introduced to remove, which makes the no-attention GRU the natural reference point. Dropout of 0.3 is applied at the GRU input and output; a final linear layer projects to vocabulary logits.

### D. Attention decoders

The three attention decoders share a common backbone: the encoder features are linearly projected to the hidden dimension, the pooled projection initializes the GRU state, and at every decoding step a context vector formed by attending over the 49 projected locations is concatenated with the word embedding as GRU input. The variants differ only in how the attention scores are computed.

- **Luong (dot-product)** [5]: the score for each location is the dot product between the GRU hidden state  $h$  and the projected feature,  $\text{score}_n = h^\top p_n$ , followed by a softmax. The scores are *not* scaled by the hidden dimension.
- **Bahdanau (additive)** [4]: the score is  $v^\top \tanh(W h + U p_n)$  with learned  $W$ ,  $U$ , and  $v$ , followed by a softmax.
- **Multi-head** [6]: scaled dot-product attention with four heads, each computing  $\text{softmax}(q k^\top / \sqrt{d_{\text{head}}})$  over the locations, with learned query/key/value/output projections.

### E. Transformer decoder

The Transformer decoder [6] replaces the GRU entirely: two decoder layers with four attention heads, model dimension 512, feed-forward dimension 1024, learned positional embeddings, causal self-attention over the caption prefix, and cross-attention over the 49 projected encoder features. At 10.0M trainable parameters it is the largest model in the comparison, roughly  $2.4 \times$  the baseline.

### F. Shared decoder interface

To make the comparison mechanically fair, every decoder implements the same interface: a teacher-forced forward pass returning per-token logits, and an incremental single-step decode used identically at evaluation. The recurrent attention models additionally expose their learned per-step spatial attention weights (shape [batch, heads, 49]) for visualization; the baseline exposes none, and our Transformer implementation does not expose the native internal cross-attention matrices, so Transformer attention maps are not included in the qualitative analysis. Because the harness calls this one interface, the training loop, decoding, scoring, and logging are identical for all five models, removing a common source of accidental discrepancy.

## III. DATASET AND TRAINING SETUP

### A. Dataset

We use Flickr8k [9], each image annotated with five human-written captions. Images and captions were obtained from the Jason Brownlee Flickr8k release mirror, and the caption-split definition from the Stanford Karpathy archive [10]. The downloaded archive contains 8,091 image files; the Karpathy definition assigns 8,000 of them to 6,000 training, 1,000 validation, and 1,000 test splits, and the remaining 91 files are unused. Captions are lowercased and tokenized; the vocabulary is built from the *training* captions only, keeping words that occur at least five times, yielding 2,541 tokens including `<start>`, `<end>`, `<pad>`, and `<unk>`. Pairing each image with each of its five captions gives 30,000 training image-caption pairs.

### B. Data-integrity safeguards

The data definition (split assignment, tokenized captions, vocabulary, and canonical image ordering) is built once and content-hashed with SHA-256. The cached encoder features are bound to a hash of the image ordering, so a feature cache built from a different image order—which would otherwise have the identical tensor shape and silently mispair every image with another image’s features—is detected and rejected. Every checkpoint and results file records these hashes, and the final evaluation refuses to run unless they match. These safeguards matter because the two authors trained on separate machines and merged results; the hashes guarantee all models were trained against one data definition. The frozen run manifest confirms identical `vocab`, `splits`, and `file_order` hashes across all five architectures.

### C. Training procedure

All models use the Adam optimizer [11], cross-entropy loss with teacher forcing, gradient-norm clipping at 5.0, batch size 64, and a maximum of 30 epochs with early stopping (patience 6) on validation BLEU-4. Padding tokens are masked from the loss. The GRU models use 256-dimensional word embeddings and 512-dimensional hidden states; the Transformer uses model dimension 512 and feed-forward dimension 1024. Dropout 0.3, the batch size, epoch budget, gradient clipping, data processing, and evaluation procedures are shared across all models, and each architecture receives the same three-value learning-rate search budget. These choices were fixed *before* any model was implemented, so a slower-converging architecture cannot be disadvantaged by a stopping rule chosen after seeing its behavior. Training ran on a single NVIDIA T4 GPU in Google Colab.

### D. Experimental protocol

The protocol is designed so the final comparison is trustworthy:

- 1) **Per-architecture learning-rate search.** Each model is trained once at each of three learning rates ( $1 \times 10^{-4}$ ,  $3 \times 10^{-4}$ ,  $1 \times 10^{-3}$ ) using a dedicated search seed, selecting the best by validation BLEU-4. A shared learning rate would not be equally fair to architectures with different optimization behavior; a small per-architecture search is the compromise. The four GRU models selected  $3 \times 10^{-4}$ ; the Transformer selected  $1 \times 10^{-4}$ .
- 2) **Three independent seeds.** At its selected rate, each model is retrained under three fresh seeds. The search run is *not* reused, since its score was optimistically selected. We report mean and standard deviation over the three seeds, and—using the same three seeds for every model—compare models pairwise seed-by-seed.
- 3) **Validation-only selection, gated test.** Learning rates and checkpoints are chosen solely on validation BLEU-4. The test set is evaluated once, only after every decision was frozen into a manifest that records the exact checkpoints, hashes, and selection rule.

*Disclosure.* During initial pipeline development, before the protocol was finalized, preliminary test-set scores were computed for the baseline. After the protocol was adopted, no further test evaluation occurred until all decisions were frozen. The test set was therefore not entirely unseen during early baseline development, and we report final test numbers with that caveat.

### E. Evaluation metrics

We compute BLEU-1 through BLEU-4 [12] using NLTK corpus BLEU (brevity penalty applied, no smoothing), a single implementation identical for every model, together with METEOR [13] and a custom TF-IDF cosine-overlap score, all at the corpus level against all five reference captions per image. Table I reports BLEU-4 as the primary comparison metric; the complete BLEU-1 through BLEU-3 values are computed identically and included with the repository artifacts.

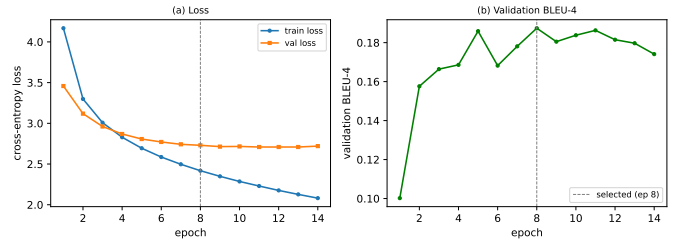


Fig. 1. Baseline training dynamics (seed 1). (a) Training and validation cross-entropy loss; (b) validation BLEU-4. The dashed line marks the selected epoch. Validation BLEU-4 sits on a noisy plateau, motivating checkpoint selection on BLEU-4 rather than loss.

Two caveats on the metrics. First, the TF-IDF caption-overlap score is inspired by CIDEr [14] but is *not* standard CIDEr or CIDEr-D—it treats each reference caption as a document, merges n-gram orders into one vector, and omits CIDEr-D’s clipping, length term, and scaling—so its values are small and meaningful only for relative comparison among our models, not as absolute scores. Second, all scores are intended for *internal* comparison; because published captioning results differ in tokenization, vocabulary, decoding, and metric implementation, we do not treat our numbers as directly comparable to them.

## IV. RESULTS AND ANALYSIS

### A. Baseline stability and training behavior

Fig. 1 shows the baseline’s loss and validation BLEU-4 for one seed. Training loss decreases monotonically while validation loss flattens and ticks up after roughly epoch 11—mild overfitting. Validation BLEU-4 sits on a noisy plateau from about epoch 5, and the two selection criteria disagree about the best epoch (validation loss reaches a broad minimum around epochs 9–13, while validation BLEU-4 peaks at epoch 8). This is why we select checkpoints on BLEU-4, the quantity we report, rather than on loss.

### B. The validation and test sets tell different stories

On the *validation* set the baseline’s three seeds were extraordinarily tight (BLEU-4 spread of 0.0004), which made every gap between models look decisive. On the *test* set, the baseline’s own three seeds span 0.0144 (Table I); the validation range was 0.00037 and the test range 0.0144, roughly 38 times larger. This is the single most important methodological observation in our results: the apparent precision of a single-split comparison is misleading, and differences small relative to the observed seed-to-seed variation should be interpreted cautiously. We therefore treat the held-out test numbers as primary and report per-seed spread rather than bare means. Had we run a single seed, or reported only validation, we would have drawn substantially over-confident conclusions.

### C. Only additive attention improves mean test BLEU-4

Table I gives the full comparison. Reading it against the baseline’s test seed spread of 0.0144, the picture is:

TABLE I

FIVE-MODEL COMPARISON. ALL VALUES ARE MEAN OVER THREE SEEDS (POPULATION STANDARD DEVIATION ACROSS THE THREE RUNS IN PARENTHESES FOR THE VALIDATION AND TEST BLEU-4 COLUMNS). TEST BLEU-4, METEOR, AND TF-IDF OVERLAP ARE ON THE HELD-OUT TEST SET; “TF-IDF OVERLAP” IS OUR CUSTOM CIDER-INSPIRED SCORE, NOT STANDARD CIDER. “PAIRED” IS THE MEAN PER-SEED TEST BLEU-4 DIFFERENCE FROM THE BASELINE USING MATCHED SEEDS; “CONSISTENT” MEANS THE SIGN AGREED ON ALL THREE SEEDS. LATENCY IS AMORTIZED GREEDY DECODING AT BATCH 64, MAX LENGTH 30, ON A T4. MODEL SELECTION: MAX VALIDATION BLEU-4.

| Model                     | Params | Val BLEU-4      | Test BLEU-4            | METEOR        | TF-IDF overlap | ms/image   | Paired vs. baseline  |
|---------------------------|--------|-----------------|------------------------|---------------|----------------|------------|----------------------|
| GRU + Bahdanau (additive) | 5.76M  | 0.1974 (0.0028) | <b>0.2062</b> (0.0063) | 0.4115        | 0.0413         | 0.7        | +0.0074 (mixed)      |
| GRU baseline (no attn.)   | 4.19M  | 0.1873 (0.0002) | 0.1987 (0.0059)        | <b>0.4150</b> | 0.0403         | <b>0.4</b> | —                    |
| Transformer               | 10.03M | 0.1959 (0.0023) | 0.1973 (0.0045)        | 0.4033        | 0.0389         | 2.6        | −0.0014 (mixed)      |
| GRU + multi-head          | 6.29M  | 0.1863 (0.0014) | 0.1970 (0.0042)        | 0.3985        | 0.0382         | 0.8        | −0.0018 (mixed)      |
| GRU + Luong (dot-product) | 5.23M  | 0.1735 (0.0007) | 0.1798 (0.0052)        | 0.3850        | 0.0350         | 0.6        | −0.0189 (consistent) |

- **Bahdanau (additive) attention is the top performer** on both validation and test (test BLEU-4 0.206), and leads on TF-IDF overlap, though the baseline remains slightly higher on METEOR (0.4150 vs. 0.4115). Its per-seed test advantage over the baseline is *mixed*: it wins on two seeds, but the strongest baseline seed (0.2055) edges the weakest Bahdanau seed (0.1975). The defensible claim is that Bahdanau is the top performer and the only attention mechanism to improve on the baseline in mean BLEU-4, with a margin comparable to seed variance at  $n = 3$ .
- **Luong (dot-product) attention produced lower test BLEU-4** than the baseline under all three matched seeds, by a mean of 0.019. It is the only architecture whose paired difference had the same (negative) sign on all three seeds, although three runs are insufficient for statistical significance.
- **Multi-head attention and the Transformer showed no consistent advantage over the baseline**: both have mixed per-seed signs and near-zero mean differences. Notably the 10.0M-parameter Transformer does not beat the 4.2M baseline, and additive attention at 5.8M outperforms it in mean BLEU-4.

The METEOR and TF-IDF-overlap orderings broadly track BLEU-4 (Bahdanau and the baseline lead, Luong trails), which indicates the ranking is not an artifact of BLEU alone. The one crossover is that the baseline edges Bahdanau on METEOR while trailing on BLEU-4 and TF-IDF overlap, consistent with the two models being close.

The headline is therefore stronger than “attention helps.” On mean test BLEU-4, three of four attention mechanisms fail to improve on a no-attention baseline, one produces substantially lower mean BLEU-4, and the only gain comes from the simplest additive-attention variant rather than the Transformer. This is consistent with the hypothesis in Section I: a strong frozen ResNet-50 representation already captures much of what spatial attention would otherwise recover, shrinking or erasing its benefit on a small dataset.

#### D. A plausible failure mode for dot-product attention

Fig. 2 visualizes per-word attention for the best and worst models on one example image. Bahdanau attention varies across generated words and distributes mass over the dog and several surrounding water regions. Luong attention, by

contrast, after the first word repeatedly concentrates on nearly the same lower-right location for the remaining content words. On this example the Bahdanau model generates “jumping” while Luong generates “swimming.”

This behavior—one model’s attention shifting across words while the other’s returns to a single region—is consistent with *attention collapse* in the Luong case: a context vector that is nearly constant and carries little image-conditioned signal would leave the model paying a parameter cost for a mechanism that is effectively disabled. Unscaled high-dimensional dot products are one plausible contributor—Luong here uses unscaled dot products over 512-dimensional vectors, which produce large-magnitude scores that can saturate the softmax onto whichever location dominates early; this is the saturation that scaled dot-product attention avoids by dividing by  $\sqrt{d}$  [6], which our multi-head variant does and our Luong variant does not. We show this as a single illustrative failure case, not as proof of the architecture-wide cause; confirming that mechanism would require aggregate attention statistics across images and seeds or a scaled-single-head Luong ablation, which we leave to future work. Each panel is independently min-max normalized per word, which aids visual contrast but prevents comparing absolute attention concentration between panels.

#### E. Latency and decoding policy

Latency in Table I is greedy decoding on a T4. The four GRU models decode incrementally by construction (recurrent-state reuse) and are fast, 0.4–0.8 ms/image. The Transformer is 2.6 ms/image, but this figure reflects its decoding *policy* rather than an intrinsic architectural cost: our Transformer uses prefix recomputation (non-incremental), re-encoding the growing caption prefix at every step. A key/value-cached implementation would reduce this substantially. The latency gap should therefore not be read as “Transformers are inherently slower”; it compares an incremental GRU against an uncached Transformer, and we report it as such.

#### F. Qualitative failure modes

An automated failure scan over the 1,000 validation images surfaced a difference in *how* the baseline and the best model fail. The scan flags three patterns: degenerate-length captions, consecutive-token repetition, and a high proportion of non-function words absent from all five reference captions. The last

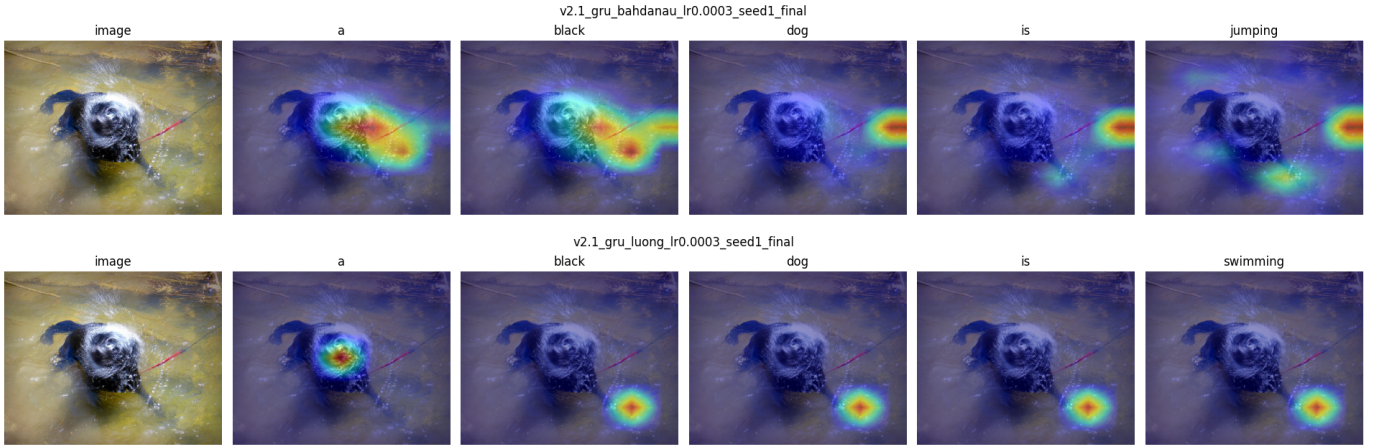


Fig. 2. Per-word spatial attention on the same validation image, Bahdanau (top) and Luong (bottom). Bahdanau attention varies across words and image regions, whereas Luong repeatedly concentrates on nearly the same lower-right region after the first generated word. The Bahdanau model generates “jumping” and the Luong model “swimming.” Each panel is independently min-max normalized for visibility, so absolute intensities are not comparable across panels.

is a noisy hallucination proxy—a reference-novelty heuristic rather than out-of-vocabulary detection, since a generated word can be a legitimate synonym and still be absent from the five references. The baseline’s errors are dominated by *genericness*: it describes the overall scene while missing the specific action—e.g. “two dogs are running through a grassy area” for an image whose reference is a dog leaping to catch a ball. Qualitative inspection of Bahdanau’s captions additionally found *repeated phrases* in some outputs, such as “a blue shirt” appearing several times in one generated sentence (“a woman with a blue mohawk is wearing a blue shirt and a blue shirt is wearing a blue shirt”). Both models emit `<unk>` on rare objects. The tentative takeaway is that when attention adds specificity it can also introduce a repetition failure mode less evident in the blunter baseline—a qualitative observation rather than a quantified rate.

### G. Scope and limitations

Several analyses named in our proposal were omitted for time and compute reasons, and we note them rather than drop them silently. The encoder fine-tuning ablation was not run: all reported results use a frozen ResNet-50, chosen because the primary five-decoder comparison and the three-seed protocol exhausted the available budget. A greedy-versus-beam decoding comparison (beam sizes 3 and 5) was not run either; beam-search code is included in the repository but all reported numbers use greedy decoding. The full all-model qualitative caption panel and per-architecture attention maps for the multi-head and Transformer decoders were also not produced; we show the Bahdanau and Luong maps as an illustrative contrast only. Finally, our TF-IDF overlap score substitutes for standard CIDEr-D, which we did not implement. None of these omissions affects the quantitative five-model BLEU/METEOR comparison, which is complete.

## V. LESSONS LEARNED

**A strong frozen encoder reshapes the question.** Our central finding is that decoder attention, which is decisive in the classic captioning literature built on weaker features, largely did not pay off here: only additive attention beat a no-attention baseline on test data, and the Transformer did not. With a modern ImageNet-pretrained ResNet-50 already providing a rich global representation, the marginal value of spatial attention on a small dataset is small—exactly the hypothesis we set out to test.

**Evaluate on held-out data, and measure seed variability.** The baseline’s observed seed-to-seed variability on the fixed test split (range 0.0144) was far larger than the spread of its selected validation maxima (range 0.00037). Reporting validation alone, or a single seed, would have produced confident but unsupported rankings. Running three seeds and reading them against the test-set spread turned “model A scored higher” into a claim we can actually qualify, and forced us to label most differences as within seed variability rather than decisive. We note this measures training-seed variability on one fixed split, not test-sample or dataset-split uncertainty.

**Visualization can motivate an explanation, not just decorate it.** The Luong result was not merely reported as a lower score; the attention heatmap showed a concrete failure mode on one example—concentrating on a fixed off-object location—which pointed us toward a plausible cause (unscaled dot-product saturation) to investigate, rather than leaving the deficit unexamined. Establishing that cause across the model would require aggregate statistics or a controlled ablation we did not run.

**Selection metric and test discipline matter.** Validation loss and validation BLEU-4 peaked at different epochs, so selecting on loss would have reported a different, worse model on the metric we care about. Selecting on the reported metric, and gating the single test evaluation on a frozen manifest, kept the comparison honest.



**Reproducibility infrastructure earns its keep.** Because we trained on separate machines and merged results, mismatches—a differently ordered feature cache, a stale result from a since-fixed model, a checkpoint saved under a changed library default—were real risks. Content-hashing the data definition, versioning runs, and gating evaluation on a frozen manifest caught these mechanically; the manifest’s matching hashes across all five models are our evidence that the comparison is valid. A concrete instance: a mid-project PyTorch change to the default `weights_only` argument of `torch.load` broke checkpoint resumption until the loader and the version metadata were corrected—exactly the kind of silent, environment-dependent failure the hashing and versioning were meant to surface.

#### CONTRIBUTIONS

Anthony implemented the data pipeline, the frozen ResNet-50 encoder and feature caching, the shared training harness and experimental protocol, and the no-attention GRU baseline. Aadit implemented the Luong, Bahdanau, multi-head, and Transformer decoders, the METEOR and TF-IDF-overlap evaluation and beam search, and the attention visualizations and failure analysis. Both authors jointly tuned hyperparameters, analyzed results, and wrote this report.

#### CODE AVAILABILITY

The complete code, experiment definitions, frozen run manifest, and instructions to reproduce these results are available in the `Final Project` directory of:  
<https://github.com/Dokkae-bi/Intro-to-DL-Summer-26>

#### REFERENCES

- [1] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.
- [2] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, “ImageNet: A large-scale hierarchical image database,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2009, pp. 248–255.
- [3] K. Cho *et al.*, “Learning phrase representations using RNN encoder-decoder for statistical machine translation,” in *Proc. Conf. Empir. Methods Nat. Lang. Process. (EMNLP)*, 2014, pp. 1724–1734.
- [4] D. Bahdanau, K. Cho, and Y. Bengio, “Neural machine translation by jointly learning to align and translate,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.
- [5] M.-T. Luong, H. Pham, and C. D. Manning, “Effective approaches to attention-based neural machine translation,” in *Proc. Conf. Empir. Methods Nat. Lang. Process. (EMNLP)*, 2015, pp. 1412–1421.
- [6] A. Vaswani *et al.*, “Attention is all you need,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 5998–6008.
- [7] K. Xu *et al.*, “Show, attend and tell: Neural image caption generation with visual attention,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2015, pp. 2048–2057.
- [8] O. Vinyals, A. Toshev, S. Bengio, and D. Erhan, “Show and tell: A neural image caption generator,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 3156–3164.
- [9] M. Hodosh, P. Young, and J. Hockenmaier, “Framing image description as a ranking task: Data, models and evaluation metrics,” *J. Artif. Intell. Res.*, vol. 47, pp. 853–899, 2013.
- [10] A. Karpathy and L. Fei-Fei, “Deep visual-semantic alignments for generating image descriptions,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 3128–3137.
- [11] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.
- [12] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: A method for automatic evaluation of machine translation,” in *Proc. 40th Annu. Meeting Assoc. Comput. Linguist. (ACL)*, 2002, pp. 311–318.
- [13] S. Banerjee and A. Lavie, “METEOR: An automatic metric for MT evaluation with improved correlation with human judgments,” in *Proc. ACL Workshop Intrinsic Extrinsic Eval. Meas. Mach. Transl. Summ.*, 2005, pp. 65–72.
- [14] R. Vedantam, C. L. Zitnick, and D. Parikh, “CIDEr: Consensus-based image description evaluation,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 4566–4575.